

DEPARTMENT OF CHEMICAL ENGINEERING IIT KHARAGPUR
END SPRING SEMESTER EXAMINATION 2010

Subject: Instability & Pattern Formation in Thin Polymer Films (INSTABILITY)

Subject

No: CH 62052

No of Students: 30

FULL MARKS: 60

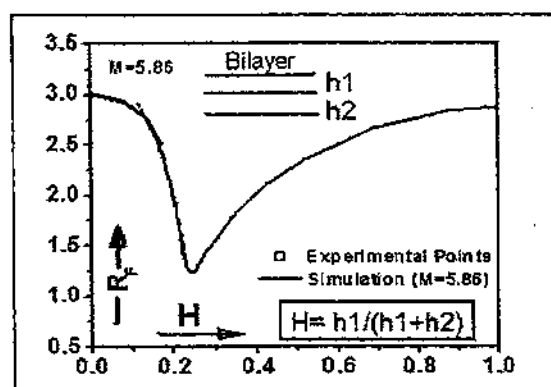
TIME: 3 Hours

Closed Book, Closed Notes

General Instructions:

1. All Questions are compulsory.
2. Clearly write your name, Roll No., Subject Name (INSTABILITY), Subject Number on the Answer Book.
3. Feel free to assume any missing data with proper justifications.
4. **Be Precise with your answers. Long, redundant answers can potentially fetch zero!**

1. (i) What is Elastic Contact Instability (ECI)? [1]
 (ii) What are the morphological evolution sequences during bonding phase in ECI? [1]
 (iii) How does the length scale of the ECI patterns (λ) scale with film thickness (h)? [1]
 (iv) How does the above scaling relation change as a function of contactor surface tension? [1]
 (v) Under what condition does one observe the "Array of hole formation" in Elastic Contact Lithography (ECL) with a 1-D stripe patterned stamp?



Explain with reference to a 1 – D striped stamp with periodicity of $3.15 \mu\text{m}$. [2]

- You are performing ECL with the above stamp (periodicity of $3.15 \mu\text{m}$). What are the likely morphologies when film thickness is (A) $0.975 \mu\text{m}$ and (B) $3.93 \mu\text{m}$. Please answer with appropriate sketches. [1+2]
- Now you are performing ECL with the same stamp, but with an **ELASTIC BILAYER** having $M=5.86$ (ratio of shear moduli of the two layers). With the same $3.15 \mu\text{m}$ periodicity 1-D striped stamp, you observe the "array of hole formation". What are the thicknesses of the individual layers? The total thickness of the bilayer is $1.4 \mu\text{m}$. Please use the attached graph for calculations. [3]

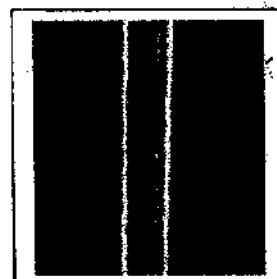
[Total for Q 1: 1+1+1+1+2+(1+2)+3 = 12]

2. (i) What are the possible modes in the dewetting of a polymer bilayer? [1]
 (ii) Discuss the likely morphological evolution sequence with neat sketches for the different modes in a qualitative manner. [4]
 (iii) You have a dewetting bilayer comprising of two films of identical thickness. If the attractive interaction between the two interfaces [(film – air) – (film – film) and (film – film) – (film – substrate)] are also identical in magnitude, which mode of deformation would you expect to prevail? Why? [2]
 (iv) Does the answer to the previous question in any way depend on the viscosity of the individual films? Justify your answer. [2]

- (v) How does the nature of initial film conformation with respect to a patterned substrate dictate the final dewetted morphology on a topographically patterned substrate? Give your answer with the example of dewetting on a stripe patterned substrate. [2]
- (vi) What happens if the stripes become progressively narrower? [1]
- (vii) What happens if the stripes become progressively wider? [2]

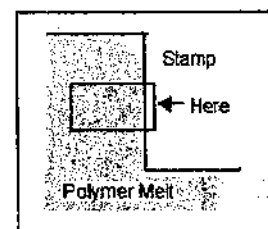
[Total for Q 2: 1+4+2+2+2+1+2 = 14]

- 3. (i) What are the major modes in an Atomic Force Microscope? [1]
- (ii) Which mode you would like to use for soft polymeric films? Why? [2]
- (iii) As can be seen in the adjacent figure, a polymer film has been scratched with an AFM tip. Under what mode this scratching has been performed? Any other specific condition which is necessary for scratching, as compared to the normal operation of an AFM? [2]
- (iv) Discuss (with sketches as necessary) the sequence of approach in an AFM for any mode of your choice. Your discussion can start from the point that the tip is aligned with a zero volt reading on the PSD. [3]
- (v) What is the smallest lateral dimension you can resolve with an AFM? [1]
- (vi) What mode would you use to image a micro contact printed surface? Why? [2]



[Total for Q 3: 1+2+2+3+1+2 = 10]

- 4. (i) Draw a qualitative velocity profile of the polymer melt during Nano Imprint Lithography. [2]
- (ii) How do stresses get accumulated during NIL? [1]
- (iii) In case there is slippage on the surface of the stamp (in NIL) will it have any influence on the pattern replication? (Help/ Make things worse/ No effect). Please discuss in a couple of sentences. [2]



[Total for Q 4: 2+1+2=5]

- 5. (i) For a thin film of thickness h coated on a semi-infinite thick substrate, find the expression for excess free energy based on van der waal's interaction (ΔG^{LW}), in terms of effective Hamaker constant (A_E). [5]
- (ii) Write down the expression for spreading coefficient (S_{21}) [1]
- (iii) Discuss film stability and instability from the stand point of Spreading Coefficient. [2]
- (iii) Express Equilibrium Contact angle in terms of S_{21} [2] [Q 5 Total: 5+1+2+2=10]

- 6. (i) What does an "UV – 400" Tag on a good quality commercial Sun Glass Means? [1]
- (ii) Why a chewing gum does not stick inside your mouth, but sticks immediately once it is outside? [2]
- (iii) Can the equations of a Boundary Layer be used for describing the Hydrodynamics of a free surface? [2]
- (iv) Write couple of advantages and disadvantages of Instability mediated patterning as compared to normal top down lithography techniques. [2]
- (v) Give examples (1 each) where surface tension has a stabilizing and destabilizing influence respectively [2] [Q 6 Total: 1+2+2+2+2=9]

All the best ☺

9. Pick up the correct answer:

- (a) MEUF (i) can be, (ii) cannot be used to treat textile effluent for dye removal.
- (b) Cloud point extraction can be done using (i) ionic, (ii) non-ionic, (iii) zwitterionic surfactants.
- (c) Cloud point of TX-100 is (i) more (ii) less (iii) equal to that of TX-114.
- (d) Reverse osmosis is (i) an equilibrium governed separation process (ii) rate governed process, (iii) none of these.
- (e) When a concentrated liquid is replaced by a dilute solution, we get (i) a shock wave, (ii) diffused wave (iii) z wave at the outlet
- (f) Electric field enhanced UF can be applied to separate (i) gelatin, (ii) PEG solution (iii) dextran solution.
- (g) If operating pH is above isoelectric point of a charged particle, it becomes (i) positively (ii) negatively charged (iii) neutral.
- (h) An ideal Gel controlled filtration is (i) mass transfer controlling (ii) pressure controlling
- (i) Osmotic pressure is a (i) colligative property (ii) not a colligative property.
- (j) Osmotic pressure of salt is a (i) linear (ii) non-linear function of concentration. (10)

10. In a gel layer controlling ultrafiltration, the specific gel layer resistance is given as, $\alpha = \alpha_0 (\Delta P)^{0.3}$, where, α is in m/kg, $\alpha_0 = 2.56 \times 10^{16}$ and ΔP is in Pa. Gel layer thickness is given as $L = L_0 (\Delta P)^{0.8}$, where, $L_0 = 5.74 \times 10^{-11}$ and ΔP is in Pa. Gel layer density is 1050 kg/m^3 and gel layer porosity is 0.7. Membrane permeability is $5 \times 10^{-11} \text{ m/Pa.s}$. Find the value of limiting operating pressure and what are the values of flux and gel layer resistance at this pressure?

(10)

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Best wishes and I hope this course will be useful in your future research